[Column] 3 patterns of heavy snow in Hokkaido in winter

Heavy snow caused by winter-type pressure arrangement |
Heavy snow caused by developed low pressure |
Heavy snow caused by a small low pressure |

Heavy snow caused by winter-type pressure arrangement

- The cold and dry air from the continent, the warm Sea of Japan, and the topography of Hokkaido make it easier for snow to fall on the Sea of Japan side.



Figure 1 Conceptual diagram of heavy snow caused by winter-type pressure arrangement

The cold and dry air blowing from Siberia replenishes water vapor in the warm Sea of Japan and creates snow clouds.

At this time, the air in contact with the sea is warmed, but the sky above remains cold and dry, so the atmospheric state becomes unstable and snow clouds develop.

In addition, the snow clouds that have reached Hokkaido are further developed by forced rise by mountains, etc., bringing heavy snow to the Sea of Japan side, but the snow clouds are blocked by the Taisetsuzan system and the Hidaka Mountains, which are the backbone of Hokkaido, and the Okhotsk Sea and the Pacific side are sunny days. There will be more.

Examples of heavy snow caused by winter pressure arrangement

In the winter of 2011 to 2012, the winter-type pressure arrangement continued for a long time and extensive heavy snowfall in the Sea of Japan side (total snowfall from January 11 to 15)



Figure 2 Total snowfall from January 11 to 15, 2012 (Heisei 24)



Figure 3 (left) Weather map at 9:00 on January 14, 2012 (Heisei 24) and (right) image of the weather satellite Himawari

There was a developed low pressure (968hPa) in the Sea of Okhotsk, and there was a dominant Siberian high pressure system on the continent, and the winter-type pressure arrangement near Hokkaido.

In January 2012, there were many days with winter-type pressure, and the snow clouds generated on the sea, as seen in the image of sunflowers, flowed into almost the same place (the arrow in the weather map shows the approximate wind direction).

For this reason, the total amount of snowfall from January 11 to 15 was more than 1 m heavy in a line from the northern part of the Ishikari region to Yubari, but in Sapporo, which is not far away, the amount of snowfall during this period was only 6 cm.

Heavy snow caused by developed low pressure

- The area where heavy snowfall changes depending on the position of the low pressure.
- The wind blowing on the mountains (the direction of the orange arrow) is very related.
- The wind may intensify, mainly in the crowded area of the mountains, and there may also be a snowstorm.



Figure 4 Conceptual diagram of heavy snow caused by developed low pressure

Examples of heavy snow caused by developed low pressure

From January 13 to 16, 2004, record heavy snow and blizzards on the Sea of Okhotsk side due to the influence of developed low pressure



Figure 5 Total snowfall from January 13 to 16, 2004



Figure 6 (left) Weather map at 9:00 on January 14, 2004 (Heisei 16) and (right) image of the weather satellite Himawari



The developed low pressure (964hPa) above the southeast sea of Nemuro is a typical development of the low pressure system that moves eastward in the Sea of Japan and a low pressure system that moves north off the coast of Kanto near Hokkaido, and 200 to 400 hectopascals from the 13th to the 14th A or more has also developed (the arrow is the path of low pressure).

From the image of sunflowers, you can see how the developed clouds associated with the low pressure are hanging over Hokkaido (the whiter the cloud is, the more the clouds are spreading to the sky). In addition, in the Sea of Japan, there are also streaks of clouds associated with winter-type atmospheric pressure arrangement.

Heavy snow caused by a small low pressure

- When the winter-type pressure arrangement begins to loosen, a small low pressure system may occur in the western sea of Hokkaido, resulting in local heavy snow and blizzards.



Figure 7 Conceptual diagram of the generation of small low pressure

In Hokkaido, when the winter-type pressure arrangement relaxes, it will be sunny and the temperature will drop mainly inland, and cold winds blowing from inland to the sea will occur. This wind and the seasonal wind from the continent will gather in the sea just west of Hokkaido.

At this time, when the valley of pressure with strong cold air in the sky approaches, a small low pressure occurs in the western sea of Hokkaido. Although the scale of the low pressure is small, the isobar line is very crowded from the south to the west of the low pressure, and the wind strengthens and snow clouds develop.

Small low pressure occurs several times during the winter. In addition, when this low pressure enters the land, it can cause local heavy snow and blizzards.

Examples of heavy snow caused by small low pressure

On December 5, 2011, a small low pressure system occurred due to strong cold air in the sky, and local blizzards in Shiroishiruyo



Figure 8 Daily snowfall on December 5, 2011 (Heisei 23)

Figure 9 (left) Weather map at 9:00 on December 5, 2011 (Heisei 23) and (right) image of the weather satellite Himawari



As the winter-type pressure arrangement continued, a small low pressure system with strong cold air in the sky occurred on the west coast of Hokkaido and slowly moved south. This low pressure gradually became stronger as it entered the land near Ishikari Bay, and it became a winter-type pressure arrangement again.

If you look at the weather map, the isometric lines from the south to the west of this low pressure (arrow) are crowded, and in the sunflower image, you can see that the center is swirling small like the eye of a typhoon. In addition, developed snow clouds were seen from the south to the west of the vortex, and in Shin-Shinotsu, where these snow clouds passed, the daily snowfall was 54 cm, with a maximum daily wind speed of 24.2 m/s (west wind).